

The Influence of Unsportsmanlike Fouls on Basketball Teams' Performance According to Context-Related Variables

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The aim of the current study was to analyze the temporal effects that unsportsmanlike fouls may have on basketball teams' scoring performance under consideration of context-related variables. The authors analyzed 130 unsportsmanlike fouls from 362 elite basketball games (men's and women's Olympic Games, European and World Championships). The context-related variables studied were score-line, quality of opposition, timeout situation, minutes remaining, and player status. The data were analyzed with linear-regression models. The results showed that both teams (the team that made the foul and the opponent) had similar positive scoring performances during 1 and 3 ball possessions after the unsportsmanlike foul (short-term effect). However, 5 ball possessions after the foul (midterm effect), the team that made the foul had a scoring disadvantage (-0.96) and the opponent team an advantage (0.78). The context-related variable quality of opposition was significant only during 1 ball possession, with negative effects for the team that made the foul and positive effects for the opponent. The final outcome showed a positive effect for score-line when the unsportsmanlike foul was made (0.96) and for quality of opposition (0.64).

Keywords: behavioral violation, situational variables, performance analysis, coaching

Basketball is a high-scoring and fast-paced team sport that involves quick decisions and actions. These characteristics make basketball a complex and dynamic sport (both physically and psychologically) in which each team tries to destabilize the opponent's balance and in turn generate disorder, while trying to maintain its own stability and self-organization.¹⁻³ From a complex and nonlinear perspective,^{4,5} a basketball game is influenced by perturbations or critical events that may disrupt or modify the team's self-organization.^{6,7} In fact, some game events and scenarios have direct effects on the team's subsequent scoring performance.⁸ According to Ferreira et al.,⁷ the precipitating events in basketball were disciplinary interventions (ie, technical or unsportsmanlike fouls), intentional game interruptions (ie, timeouts or substitutions), player limitations (ie, 4th or 5th personal foul), and technical and tactical incidences (ie, defensive system variations). Pertinent to the present research, some events are more important than others, such as unsportsmanlike or technical fouls. Of further relevance, previous research has suggested that game criticality in basketball is dependent on previous and following events during a game.⁷ Bar-Eli et al.⁹ argued that the criticality level of each situation increased from the beginning to the last phase of a game without linear trends. In fact, the fourth game quarter is the most critical in a game as it decides the final outcome. Therefore, coaches should be aware of the impact of these events and try to anticipate (or modify) the critical moments that may occur during the different phases of the game to maintain performance stability in their own team and/or potentially induce disorder in the opponent team.⁸

One of the basketball rules related to major violations as a critical incident is the unsportsmanlike foul. This foul is considered a violent action or rough play when the player does not show any effort to play the ball when the contact occurs, the player causes an excessive foul when he or she tries to play the ball, and a defensive player tries to stop a fast break (causes contact) and there is no opponent between the opponents' basket and the offensive player. Unsportsmanlike fouls are followed by free throws and a subsequent throw-in for the nonfouled team (International Basketball Federation [FIBA], 2014, rule 36.1).

Varca¹⁰ classified basketball players' behaviors during games according to the arousal states that they display in assertive or aggressive situations. Assertive behaviors were described as being associated with intensity and emotion but still playing within the regulations (ie, a regular foul trying to steal the ball). Aggressive behaviors were described as any behavior that was intended to directly harm or injure a player.¹¹ The aggressive behaviors are classified as instrumental, when they facilitated goal attainment when completing a desired task performance (ie, impulsive and hard defensive pressure for blocking a shot or stealing the ball), or hostile aggression, behaviors directly related to violence (ie, fouling the opponent player, pushing him or her away with a hard contact and no intention of playing the ball).^{11,12} In line with this distinction, Bar-Eli et al.¹² wrote that 2 components are necessary when a player makes an unsportsmanlike foul: intentionality and brutality. Both are characteristic of basketball that involves body contact between players from confronting teams. According to the psychological performance-crisis theory of Bar-Eli et al.,⁹ these behavioral violations can be considered game-criticality indicators. More specifically and of particular importance to the present research, it is assumed that when players fall into a psychological crisis this can coincide with behavioral violations.¹³ In support of this line of reasoning, research has shown that when players are confronted with extreme conditions (eg, game pressure or crowd influence), these violations are more frequent and intense than during normal competitions.^{6,13,14}

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On the other hand, from a coach's perspective, unsportsmanlike fouls can be understood as intentional fouls that can be used in different strategic (eg, to stop the game time) and tactical situations (eg, to stop an opponent's fast-break situation) with the aim to win the game. Indeed, these actions can avoid field goals by the opponents and send the player to the free-throw line.^{15,16} Therefore, it is important to distinguish between major violations that indicate a player's crisis and major violations that are intentionally used during tactical or strategic situations.¹³

Kirker et al¹⁷ stated that aggression in basketball, such as unsportsmanlike fouls, is the result of a combination of factors that happen before (eg, lost possession or missed shot, poor scoring performance, score-line differences within 10 points or game criticality) and after the violation (eg, better performance by the team that received the foul than the fouling team). In addition, the game importance may affect the violations, with fewer unsportsmanlike fouls the more critical the game is (eg, knockout stage). In addition, the game period and score-line are factors closely related to aggression and unsportsmanlike fouls.¹⁷ The players are more vulnerable to crisis during the last phase of a game in which time pressure, fatigue, or poor performance may lead to increased violations and in turn a psychological performance crisis.^{6,13,18} Scoring performance is influenced by a complex interplay of all kinds of factors in basketball and therefore needs to be investigated from a multidimensional perspective that integrates contextual, psychological, and cognitive factors.^{19,20} Hence, unsportsmanlike fouls are investigated in their natural environment (high-order interactions) in the present research, where all the important events are present (context-related factors) and the previous and following ball possessions can be taken into account (time-dependent effects for successful or unsuccessful actions).^{9,17}

This approach would give new insights into the impact of unsportsmanlike fouls during basketball games. Thus, the use of multivariate linear-regression analyses can be considered a useful approach for identifying and predicting the normative profiles of unsportsmanlike fouls in complex contexts.^{21–24} Therefore, the aim of this study was to analyze the temporal effects (short-, mid-, and long-term effect) that unsportsmanlike fouls may have on basketball scoring performance and final outcome in relation to context-related variables for the fouling team and the opponent. In general, we hypothesized that the team that made the unsportsmanlike foul would have lower scoring performance after the foul than the opponent team and this situation would be affected by context-related variables.

Methods

Sample

The study analyzed all the games from the 2010 World Championship ($n = 80$ and $n = 62$), the 2011 European Championship ($n = 90$ and $n = 54$), and the 2012 Olympic Games ($n = 38$ and $n = 38$) for men's and women's basketball teams, respectively, in which a total of 362 games were played. The final sample was composed of 130 unsportsmanlike fouls (men's teams $n = 79$, 0.38 unsportsmanlike fouls per game; women's teams $n = 52$, 0.34 unsportsmanlike fouls per game) gathered from the official play-by-play box scores from the FIBA official open-access Web domain (www.fiba.com). The games that included an unsportsmanlike foul ($n = 124$) showed mean score difference between confronting teams at the end of the games and when the fouls were committed of 12.8 ± 9.5 (95% CI

$= 11.17–14.43$) and 8.9 ± 8.7 (95% CI $= 7.4–10.4$), respectively. The local institutional review board approved this work.

Procedures

The data were gathered by professional statisticians from the FIBA; however, a subsample of 26 unsportsmanlike fouls (20% of the sample) were randomly selected and observed by 2 experienced analysts (basketball coaches with more than 10 y of experience). The results showed coefficients of agreement of 1.0 for points scored and received by both teams in each unsportsmanlike foul situation (previous and following 1, 3, and 5 ball possessions and free throws awarded).

All the fouls were committed during the group stage, and none occurred during the knockout stages. The unsportsmanlike fouls occurred during field-goal situations (2-point field goals $n = 74$, 56.9%; 3-point field goals $n = 29$, 22.3%), fouls with bonus situations ($n = 9$, 6.9%), after a ball was recovered ($n = 8$, 6.2%), turnovers ($n = 6$, 4.6%), or rule violations ($n = 4$, 3.1%).

To analyze the temporal effects that unsportsmanlike fouls had on basketball scoring performance, the points scored and received in the previous and following 1, 3, and 5 ball possessions and from foul free throws were studied. According to Ferreira et al¹⁷ the ball possession should be used as a temporal parameter (ie, length) and the point differences after the free throw awarded by the unsportsmanlike foul as the performance parameter (ie, criticality measure) when analyzing critical incidents. This procedure allows analyzing the potential short-term effects (immediate) when comparing the teams' point differences (points scored minus points received) between the prior first and third ball possessions and the first and third ball possession after the unsportsmanlike foul. The medium-term effects were investigated by comparing the previous and following 5 ball possessions. As a result, the total ball possessions gathered were as follows:

For the team that made the unsportsmanlike foul: $n = 130$, $n = 248$, and $n = 241$ ball possessions for the previous and following 1, 3, and 5 ball possessions, respectively

For the opponent team: $n = 130$, $n = 253$, and $n = 247$ ball possessions, respectively

Finally, the following context-related variables when the unsportsmanlike fouls were committed were studied: score-line, the point difference between confronting teams; quality of opponent, studied as the final ranking difference between the 2 confronting teams; timeout situation, studied if the fouling team called a timeout or not after the unsportsmanlike foul was committed; minutes remaining to finish the game; and player status, the role of the fouling player studied according to the player's importance on his or her team as a starter or a nonstarter player.^{19,20}

Statistical Analysis

Linear-regression analyses were used to estimate the effect of independent variables on the teams' point differences (ie, points scored – points received) for the team that made the unsportsmanlike foul and for the opponent. Two linear-regression models were applied to identify the effects of unsportsmanlike fouls in the short- and midterm (1, 3, and 5 ball possessions) and on the final outcome (final point differences). The Durbin-Watson test was used to check whether the residuals in the model were independent and to control for collinearity effects. All the models showed no heteroscedasticity in residuals or multicollinearity among regressors.

When interpreting the linear-regression results, positive or negative coefficients indicate a positive (greater) or negative (lower) propensity to increase or decrease the teams' point differences. Gender effects were tested in all the analyses (1, 3, and 5 ball possessions and final outcome); however, these analyses were excluded from all the models, as gender did not explain any additional variance.

The multiple linear-regression models for (1) short- and midterm (1, 3, and 5 ball possessions) effects and (2) the final outcome effect were defined as follows:

$$SC = \beta_0 + (\beta_1 \times UF) + (\beta_2 \times QO) + (\beta_3 \times TO) + (\beta_4 \times MinR) + (\beta_5 \times Player) + \varepsilon_i \quad (1)$$

$$FPD = \beta_0 + (\beta_1 \times SC) + (\beta_2 \times QO) + (\beta_3 \times TO) + (\beta_4 \times MinR) + (\beta_5 \times Player) + \varepsilon_i \quad (2)$$

where SC = the teams' point differences; FPD = final point differences between confronting teams; UF (unsportsmanlike foul) = a dummy variable stating whether the points were scored before (UF = 0) or after (UF = 1) the foul; QO = quality of opposition; TO = timeout as a dummy variable that controls for a timeout called by the team that made the foul (TO = 0) or not (TO = 1) after the unsportsmanlike foul was made; MinR = minutes remaining; Player = whether the fouling player is a starter (Player = 0) or a nonstarter (Player = 1); β_0 is the intercept of the regression model; $\beta_1, \beta_2, \beta_3, \beta_4$, and β_5 are the effects of the regressors; and ε_i is the disturbance term.

Effect-size calculations for a given R^2 were done using Cohen f^2 , and their interpretation was based on the following criteria: 0.02 = small effect, 0.15 = medium effect, 0.35 = large effect.²⁵ Statistical analyses were performed using SPSS 22.0 (IBM Corp, Armonk, NY, USA). The level of significance was set at $P \leq .05$.

Results

Figure 1 shows the distribution of unsportsmanlike fouls according to each game quarter. The results showed that 30% of the fouls were committed during the first half of the games (10% and 20% during the 1st and 2nd quarters, respectively) and 70.0% of the fouls were made in the second half of the games, but particularly often during the last game quarter (25.4% and 44.6% during the 3rd and 4th quarters, respectively).

Figure 2 shows means and standard deviations for the number of points scored by the teams that made the unsportsmanlike foul and by the opponent team in the 1, 3, and 5 previous and following

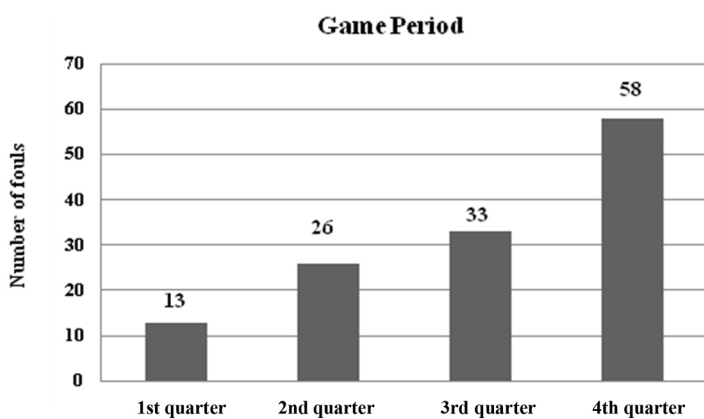


Figure 1 — Distribution of the number of unsportsmanlike fouls in the different game periods.

analyzed ball possessions of the games and in the free throws awarded by the foul.

The results for short- and midterm effects are presented in Table 1. The results of the model showed significant positive effects on points scored by the opponent team, with increments of 0.92, 0.87, and 0.78 points for the 1, 3, and 5 ball possessions, respectively. For the fouling team, the results showed positive effects on points scored, with increments of 0.93 and 0.90 points for the 1 and 3 ball possessions, respectively, but with negative effects with decrements of -0.96 points for the 5 ball possessions. The quality-of-opposition variable was only significant for the previous and following 1-ball-possession model, where a positive significant effect was obtained for the opponent team of 0.42 points. The independent variables timeout, minutes remaining, and player status did not show significant positive or negative effects on the basketball scoring performance of either teams in any of the models.

The results for final-outcome effects are presented in Table 2. The results showed significant effects of the point differences when the unsportsmanlike foul was made (positive effects of 0.96 points) and the influence of quality of opposition (positive effects of 0.64 points) when the unsportsmanlike foul was made. The independent variables minutes remaining and player status were not significant.

Discussion

The aim of this study was to analyze the temporal effects (short- and midterm and the final outcome) that unsportsmanlike fouls may have on basketball scoring performance while taking context-related variables into account. The main results showed a positive effect on scoring performance of unsportsmanlike fouls for both teams in the short term (1 and 3 ball possessions), but in the midterm and final outcome unsportsmanlike fouls only had a positive effect on the team that was fouled (ie, opponent team) and in turn a negative effect on the fouling team. These results partially support the hypothesis that after an unsportsmanlike foul the team would have lower scoring performance than the opponent team, and this situation is affected by context-related variables. In particular, major behavioral violations such as unsportsmanlike fouls are related

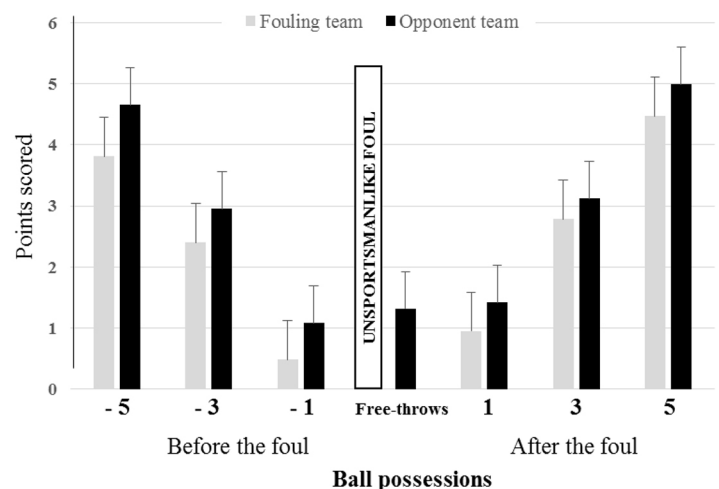


Figure 2 — Points scored by the fouling team and by the opponent team in the previous and following analyzed ball possession (5, 3, and 1 ball possessions) of the game and in the free throws awarded by the foul. The thick vertical line represents the unsportsmanlike foul (SDs are presented with thin vertical lines).

Table 1 Effects of the Independent Variables on the Point Differences According to the Number of Ball Possessions for the Fouling Team and for the Opponent Team

Variable	Ball Possessions					
	-1 / 1		-3 / 3		-5 / 5	
	Fouling team	Opponent team	Fouling team	Opponent team	Fouling team	Opponent team
Unsportsmanlike foul	0.93*** (0.51, 1.36)	0.92*** (0.50, 1.33)	0.90*** (0.47, 1.33)	0.87*** (0.46, 1.27)	-0.96*** (-4.26, 3.52)	0.78*** (0.41, 1.15)
Quality of opposition	-0.42 (-0.87, 0.03)	0.42*** (-0.03, 0.87)	-0.31 (-0.75, 0.14)	0.38 (-0.06, 0.83)	-0.25 (-0.69, 0.18)	0.35 (-0.12, 0.82)
Timeout	0.51 (-7.34, 8.36)	-0.56 (-8.41, 7.28)	4.65 (-4.26, 10.55)	-2.20 (-9.59, 6.15)	4.80 (-3.24, 10.86)	-4.68 (-13.60, 4.24)
Minutes remaining	-0.02 (-0.27, 0.23)	0.02 (-0.23, 0.27)	-0.01 (-0.27, 0.25)	-0.01 (-0.25, 0.24)	-0.02 (-0.27, 0.23)	-0.00 (-0.27, 0.26)
Player	-0.62 (-6.13, 5.07)	0.59 (-5.11, 6.28)	1.35 (-4.42, 7.12)	0.34 (-5.31, 6.10)	2.56 (-2.69, 7.81)	-1.31 (-7.14, 4.52)
Intercept	-1.48 (-7.97, 7.01)	0.64 (-7.96, 9.25)	-6.02 (-12.35, 4.30)	2.93 (-6.54, 10.40)	-3.15 (-13.25, 6.93)	5.69 (-4.97, 16.32)
Number of observations	130	130	248	253	241	247
R^2	0.37	0.38	0.42	0.38	0.50	0.43
F	5.109	5.110	5.441	5.025	6.856	5.553
P	.01**	.01**	.01**	.01**	.001***	.01**
Durbin Watson	2.22	2.20	2.09	2.28	2.38	2.09
Cohen f^2	0.59	0.61	0.72	0.61	1.00	0.75

Note: 95% confidence intervals are presented in parentheses.

** $P \leq .01$. *** $P \leq .001$.

Table 2 Effects of the Independent Variables on the Game Outcome (Final Point Differences) for the Games That Included the Unsportsmanlike Fouls

Variable	Unsportsmanlike foul ^a
Score-line unsportsmanlike foul	0.96*** (0.67, 1.25)
Quality of opposition	0.64* (0.12, 1.16)
Timeout	-1.03 (-5.68, 9.59)
Minutes remaining	-0.09 (-0.37, 0.18)
Player	0.10 (-3.37, 3.17)
Intercept	2.35 (-6.17, 10.52)
Number of observations	130
R^2	0.60
F	12.684
P	.001**
Durbin Watson	2.20
Cohen f^2	1.50

^a 95% confident intervals presented in parentheses.

* $P \leq .05$. ** $P \leq .01$. *** $P \leq .001$.

to lower performances in the mid- and long-term range, but only the contextual variable quality of opposition affected this specific situation after 1 ball possession and in the final outcome. As was argued, the unsportsmanlike foul is a critical event in basketball

that leads to free throws and a throw-in for the opponent team, which is expected to lead to a relevant advantage for the opponent team. However, this potential advantage might add additional pressure to the players and affect their performance by factors such as higher expectations, leading to stress, anxiety, and “choking” (ie, psychological crisis).^{18–20}

The unsportsmanlike-foul distribution during a game showed higher frequencies during the fourth quarter. This trend might reflect players’ being more vulnerable to crisis at the end of games.^{6,9} In fact, the fourth game quarter is the most critical as it decides the final outcome.²⁶ Therefore, it seems feasible that players are under more pressure, and, in turn, unsportsmanlike fouls (ie, major violations) occur more frequently as players are in a nonoptimal state.¹⁴ Alternatively, the higher frequency of unsportsmanlike fouls committed during the last game quarter might reflect strategic acts of instrumental aggression, because they can be considered a functional means to an end (eg, winning the game, stopping an easy field goal or a fast break, or intimidating an opponent).^{10,14,17} In this respect calling unsportsmanlike fouls can be considered an important challenge for a referee’s game management.¹⁵ During special moments of the game with a close score and a situation of high tension felt by the players, the referees might feel more inclined to call fouls that could influence the result of the game.^{15,16} As with any decision, referees might not always call unsportsmanlike fouls correctly, especially in the first half.¹⁶ Related research has shown that this is most likely because important decisions with major consequences for the game are often not awarded at the beginning of the game compared with toward the end of the game, as referees do not want to restrict their degrees of freedom. This bias of referee decision

making (which was also shown in the present analyses) poses the problem that referees are more inclined to award unsportsmanlike fouls toward the end of the game when the game is most critical.²⁷

Specifically, 79.2% of the unsportsmanlike fouls were made when shooting, 6.9% during fouls in bonus situations, and 13.9% after player mistakes. These results may indicate that unsportsmanlike fouls are actions to prevent the opponents from scoring (eg, fast-break opportunities or a clear field-goal situation) or actions related to player frustration after a mistake (ie, rule violation or a turnover). As was argued, the unsportsmanlike fouls have a psychological component that occurs before the action, such as lost possessions, low shooting effectiveness, or a close score-line^{14,17} and the coach's and players' strategies such as stopping the game pace, the opponent's scoring trend, or intimidating an opponent (ie, shooting player).^{7,17} From a psychological perspective, body contact in basketball is an inherent and necessary part of the game. Generally, coaches demand aggression during defensive actions (eg, the use of blocking or hard play to keep the offense away from the scoring zone). An important concept when interpreting aggression in basketball is the notion of moral modularity,²⁸ which suggests that aggressive play is judged and interpreted differently by offense or defense. In this respect, the defense, in general, is suggested to have a lower threshold of calling a certain action aggressive or even violent as compared with the offense. However, as the game progresses and self-regulation becomes harder due to fatigue, stress, and increasing arousal levels, this initial tendency of the defense might "spill over," leading to impaired decision making and a higher chance of committing unsportsmanlike fouls.^{6,9}

The short- and midterm effects showed similar scoring performances for both teams (the fouling team and the opponent) during 1 and 3 ball possessions; these results were not expected because unsportsmanlike fouls may generate a state of crisis or high psychological vulnerability. This may show that the fouling team deliberately tries to maintain scoring performance after the foul. One argument may be that coaches are more prepared to anticipate game perturbations, and then these fouls are used as a tactical or strategic resource.^{7,16} In fact, the short-term effects identified in the current study are contrary to crisis vulnerability that should decrease the team's performance.^{6,9,14} Specifically, the fouling team may gain a short-term psychological advantage with the unsportsmanlike foul by interrupting an opponent's positive trend. This may affect the opponent's physical and mental state, causing a brief negative momentum, and in turn may lead to a short-term positive performance boost for the fouling team.²⁹ On the other hand, the opponent team has a positive effect due to points scored by the free throws awarded (1.32 ± 0.67 points) and presumably because they might be more confident and positive after this behavioral violation when playing the next ball possession.^{14,15} However, results in 5 ball possessions showed that the fouling team has a disadvantage (-0.96 points) and the opponent an advantage (0.78 points). These trends may show that during balanced score-lines (ie, no more than 8 points difference in the present sample) the fouling team has more limitations when decreasing the score differences in a midterm period. This fact is related to crisis vulnerability, poorer performances, and the impossibility of recovering the increased differences by free throws awarded to the opponent. According to Kirker et al,¹⁷ the close score differences and game time are key factors that affect the aggression, mainly to stop a string of points by the opponents.

The final outcome effect reflected a positive influence of unsportsmanlike fouls and quality of opposition. These trends show that in general these fouls have a lower impact at the end of the game (0.96 points), but, as was stated previously, the short- and

midterm effects are more affected by this foul due to interrelated events during the game history.⁷ Thus, after the unsportsmanlike foul both teams try to keep their own stability with positive effects.¹⁻³ On the other hand, the unsportsmanlike foul is not frequent during basketball games, and then it could be used as a strategic/tactical resource to stop the game, control the game dynamics, or try to intimidate the opponent.^{7,17} In fact, the results have shown a lower impact of unsportsmanlike fouls on the game outcome. In this sense, one would expect that the unsportsmanlike foul would result in a substantial advantage for the opponent team. However, the results point out that this advantage is not as large as could be expected, potentially due to psychological and physiological aspects such as stress or anxiety, which may lead to lower performance in pressure situations where good performance is expected.^{6,9} In addition, the influence of quality of opposition during 1 ball possession and final outcome reinforces the importance of this variable in basketball during short-term periods (ie, high pressure such as free throws and the subsequent ball possession awarded by the foul) that may determine the score differences and the outcome of the game.³⁰ Thus, the identified result may reflect that the best teams perform better than low-level teams during critical situations (eg, 1 ball possession and in the overall game), trying to gain an advantage due to better decision making, tactical behaviors, or field-goal selection.³⁰

One interesting finding that emerged in the current study was the absence of a significant influence of the player variable (ie, starter or nonstarter). This may reflect that starters and nonstarters are equally affected by game pressure, stress, and psychological crisis when committing an unsportsmanlike foul.^{6,9,12} Therefore, the present findings reinforce the idea that unsportsmanlike fouls are context-dependent and not player-status-dependent.

An intriguing result is the absence of unsportsmanlike fouls during knockout games. According to Kirker et al,¹⁷ a reduced number of major violations has been reported to occur during critical games. It is probable that balanced games (ie, like the present games: scoring differences of 12 points) with a high importance for both teams (ie, international championships) may affect the players' psychological state, leading them to try to avoid major violations that generate a scoring disadvantage. In addition, during these critical games referees might be inclined to avoid high-impact decisions such as unsportsmanlike fouls¹⁴ as they do not want to create a major advantage for one team if they penalize the opponent team with an unsportsmanlike foul.¹⁵

Finally, the current study has some limitations that have to be addressed in further research. Although the present research has provided initial evidence for the potential impact that unsportsmanlike fouls can have on the fouling team, future research has to scrutinize this finding from psychological and physiological perspectives. For example, future research should control for game-location effects during national-league games, as research has highlighted this as a critical factor related to unsportsmanlike fouls.¹⁷ Further research should replicate these initial findings with different samples from national leagues (ie, regular-season and playoff stages) or different categories (ie, under 16, under 18, or under 20). In addition, some critical incidents related to critical behaviors, such as a player's fifth personal foul, technical fouls, or the fourth personal foul of an important player, are also candidate actions for future studies.

Practical Applications

The present results help illuminate the most important factors associated with unsportsmanlike fouls and effectiveness in basketball. From a practical perspective, Table 3 displays the simulated final

Table 3 Simulated Final Point Differences in the Score for the Opponent Team According To Score-Line, Timeout, Player Status, and Quality of Opposition

Score-line	Player	Timeout	Quality of Opposition					
			1	2	3	-1	-2	-3
Winning by 3 points	Starter	Yes	4	4.7	5.3	2.3	2.1	1.5
	Starter	No	5	5.7	6.4	3.8	3.1	2.5
	Nonstarter	Yes	3.9	4.6	5.2	2.7	2	1.4
	Nonstarter	No	5	5.6	6.2	3.7	3	2.4
Tied	Starter	Yes	1.2	1.8	2.4	-0.1	-0.8	-1.4
	Starter	No	2.2	2.8	3.5	0.9	0.3	0.4
	Nonstarter	Yes	1.1	1.7	2.3	-0.2	-0.9	-1.5
	Nonstarter	No	2.1	2.7	3.4	0.8	0.2	-0.4
Losing by 3 points	Starter	Yes	-1.7	-1.1	-0.4	-3	-3.6	-4.3
	Starter	No	-0.7	-0.1	0.6	-2	-2.6	-3.2
	Nonstarter	Yes	-1.8	-1.2	-0.5	-3.1	-3.7	-4.4
	Nonstarter	No	-0.8	-0.2	0.5	-2.1	-2.7	-3.4

Note: 1, 2, and 3 ranking differences: positive and negative values represent lower or better levels, respectively.

point differences in the score according to context-related variables. The simulations considered fixed values for minutes remaining (10 min). The variables player status, timeout, and quality of opposition were considered according to drawing, winning, and losing scores. The simulation results showed that if the team is winning by 3 points and receives an unsportsmanlike foul, they would win the game by a range of 6 to 1 points with a greater importance of quality of opposition (the better the opponent, the lower the final difference) and timeout called by the fouling team (lower differences if the fouling team calls a timeout). During drawing simulations, the results showed that they would win the game by 1 or 2 points if they are better than the fouling team. However, if the fouling team is better ranked than the opponent team and calls a timeout, the fouling team would win the game (by 1 point). Finally, if the team that received the foul is losing by 3 points, they would lose the game by a range of 1 to 4 points (1 exception occurred leading the team to win: if the fouling team does not call a timeout, the starter player made the foul, and the team is ranked 3 points lower than the opponent).

These game scenarios showed the importance of going from a balanced to an unbalanced situation where the coach and player decision making has extreme importance.⁷ From a coach's point of view, the proposed scenarios would give a range of information to improve game development and management of competition and game perturbations (ie, game preparation, player substitutions, or defensive options).

From a referee's perspective, the reduced impact of unsportsmanlike fouls might suggest the necessity for a rule modification. Depending on how and when an unsportsmanlike foul is committed, the fouling team may gain a benefit (see examples in Table 3). Thus, the basketball federation and referees' committee might want to consider more severe sanctions (eg, fouling player out of the court immediately after the foul to keep the game safe, the unsportsmanlike foul counts as 2 team fouls instead of 1, or avoiding the possibility for calling timeouts to the fouling team).

Finally, from a psychological point of view, this paper suggests that players have to be prepared to reduce the likelihood of unsportsmanlike fouls, especially toward the end of the game when it is the most critical and players have a harder time regulating themselves due to higher fatigue and stress. In this respect, it seems warranted to consider mental interventions before and during elite games (eg,

self-talk, breathing techniques) to avoid committing unsportsmanlike fouls and the resulting negative consequences.³⁰

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